

EEHC DISTRIBUTION MATERIALS SPECIFICATION	EDMS 18-402-1
Over-Head Accessories of Bare Connectors	14-01-2025

EDMS 18-402-1

SPECIFICATION

FOR

**OVER-HEAD ACCESSORIES OF
BARE CONDUCTORS**

Issue: January-2025 / Rev- 1

- This revision contains option items that must be selected by EDC before bidding.

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1- SCOPE

This specification describes the minimum technical requirements for the design, engineering, manufacture, testing, supply and delivery of accessories used with over-head distribution line. These accessories include: mid-span compression joint, parallel grove clamp and top tie. This required accessories shall be compatible with bare conductors that carry electrical power up to 24 kV. The required accessories shall be suitable for use in outdoor installations of the Egyptian Electricity Distribution Companies (EDCs).

2- APPLICABLE CODES and STANDARDS

This specification shall be read in conjunction with the latest revisions of the following EDMS specifications:

- EDMS 04-100 Over-head Aluminum Conductors Steel Reinforced (ACSR)
- EDMS 04-101 Over-head All Aluminum Alloy Conductors (AAAC)

The equipment/material covered in this specification shall comply with the latest edition/amendment of the following codes and standards. Where any provision of this specification differs from those of the standards listed below, the provisions of this specification shall apply.

- BS 3288 Insulator and conductor fittings for overhead power lines
- IEC 61284 Overhead lines – Requirements and tests for fittings
- DIN 48072-1 Line tapes for stranded aluminum conductors and ACSR
- BS EN 50182 Conductors for overhead lines. Round wire stranded conductors.
- BS EN 50189 Conductors for overhead lines. Zinc coated steel wires.
- BS EN 2004 Test Methods for Aluminum and Aluminum Alloy products.
- DIN EN 755-7 Aluminum and aluminum alloys - Extruded rod/bar, tube and profiles - Manufacturing tolerances.

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- IEC 61089 Round wire concentric lay overhead electrical stranded conductors.
- IEC 60888 Zinc-coated steel wires for stranded conductors.
- IEC 60889 Hard-drawn aluminum wire for overhead line conductors.
- IEC 61597 Overhead conductors – Calculation methods for stranded conductors.
- IEC 60468 Method of measurement of resistivity of metallic materials.
- IEC 61394 Overhead lines – Characteristics of greases for bare conductors.
- ISO 1461 Hot dip galvanized coatings on fabricated iron and steel articles.
- ISO 2859 Sampling procedures for inspection by attributes.

3- SERVICE CONDITIONS

The accessories are required to have a good electrical and mechanical performance under the following local service conditions:

Mean daily temperature	40 °C
Maximum ambient shade temperature	50 °C
Minimum ambient shade temperature	-5 °C
Maximum relative humidity	95 %
Environmental conditions	Humid tropical climate with heavily polluted atmosphere
Maximum Rainfall	250 mm
Altitude	Up to 1000 m above sea-level

4- DESIGN CRITERIA

4.1 General:

- Accessories shall be suitable for use on ACSR and AAAC of cross-sectional area in the range of:
 - 35/6 mm² to 240/40 mm² for ACSR conductors; and
 - 70 mm² to 300 mm² for AAAC conductors.
- The construction details and dimensions are given in the attached drawings and

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tables, which form part of this specification.

- All accessories shall be free from defects which would likely cause them to be unsatisfactory in service. All parts shall be inherently resistant to corrosion, both during storage and in service. All ferrous metal parts except those made of stainless steel shall be protected by hot dip galvanizing as per the requirement of ISO 1461.
- The electrical conductivity and current capacity of each current carrying parts shall be not less than those of the equivalent length of conductor. All accessories shall meet or exceed the performance of conductor in all respects.
- The accessory shall be able to withstand short-circuit of relatively high intensity and duration.
- Manufacturing tolerances shall be as per relevant standards. Tolerances in dimensions shall not exceed the value given in the attached drawings and tables.
- The maximum allowable decrease in weight is 3% of the specified weight.

4.2 Mid-Span Compression Joint

- Mid-span compression joint is intended to ensure mechanical and electrical connection between two lengths of conductor under mechanical strength.
- Each mid-span full tension joint comprises:
 - One aluminum alloy sleeve allowing the hexagonal compression on conductor core for AAAC conductors.
 - One steel sleeve and one aluminum alloy sleeve allowing the hexagonal compression on conductor core for ACSR conductors.
- Aluminum alloy sleeve shall be made of suitable alloy of purity not less than 99.5% for ACSR and 98% for AAAC in accordance with DIN EN 755-7.
- The joint shall have a resistivity less than 75% of the resistivity of equivalent

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length of conductor.

- The joint shall not permit slipping off, damage to or failure of the complete conductor or any part thereof at a load less than 95% of the ultimate tensile strength of the conductor.
- The steel sleeve should not crack or fail during compression. The steel sleeve shall be hot dip galvanized.
- Mid-span full tension joints shall be of regular hexagonal compression type. These joints shall guarantee mechanical and electrical features at least equivalent to those of the conductor on which they are installed.
- Joints shall be provided with a center marking for correctly positioning the conductor before crimping
- The internal faces of Aluminum alloy sleeves shall be coated with Oxide Inhibiting Grease/Compound to improve electrical contact and ensure maximum electrical performance. Tube ends should be protectively capped to prevent spoilage and spillage of the grease.
- Compression dies shall be of a common size and dimensions for a given size of conductor.
- Not more than one mid-span joint will be allowed in one span of the same conductor.
- Each individual Mid-span full tension joint shall have the range of permanent embossed markings listed below:
 - Manufacturer's identification.
 - Type of conductor
 - Cross-section of conductor
 - Compression die reference(s)
 - Reference of lot including the last two digits of manufacture year
 - Marks showing the places of compression

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- Die compression areas shall be clearly marked, and shall bear the words ‘COMPRESS FIRST’ suitably inscribed on where the compression begins. If the equipment is designed for intermittent die compressions, it shall bear the identification marks ‘COMPRESSION ZONE’ and ‘NON-COMPRESSION ZONE’ distinctly with arrow marks showing the direction of compression and knurling marks showing the end of the zones. The letters, number and other markings on finished equipment shall be distinct and legible.

4.3 Parallel-Groove Clamp

- Parallel-groove (PG) clamp of bolted type is applied to joint conductors where there is no mechanical load.
- The groove of the PG clamp shall correctly fit the conductor it is intended for use with. It shall have adequate cross sectional area and length.
- Clamp Body: The clamp body shall be made of electrolytic, high strength, corrosion resistant aluminum alloy.
- Fasteners: The minimum number of tightening bolts per clamp shall be suitable for cross-section area of conductors and at least two pieces screwed in one half of the PG clamp. Each bolt shall be fitted with one locking hex nut and two washers (flat and spring). All fasteners (bolts, nuts and washers) shall be made of hot dip galvanized steel.
- Each individual PG clamp shall have the range of permanent embossed markings listed below:
 - Manufacturer's identification.
 - Cross-section range.
 - Reference of lot including the last two digits of manufacture Year.

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4.4 Top-Tie

- The preformed ties shall be made of aluminum-clad steel compatible with AAAC and ACSR conductors in the top groove of vertical-mounted pin provide the wide application range and ensure proper installation.
- The lay direction of preformed helical rods shall be the same as that of the outer layer of the conductor to which it is applied. The preformed ties and armors shall be provided with Identification label showing the manufacturer's catalog number, insulator details and conductor diameter range.

5- TESTING

- Accessories shall be tested in accordance to the latest relevant standards and this specification. It shall be the responsibility of the manufacture to perform or to have performed all the tests specified.

5.1 Type Tests

- Type tests had to be performed by accredited laboratories. Single connector of each type can be type-tested to cover its lower sizes. The type test reports shall include:
 - Electrical type tests (resistance and heat cycling tests) in accordance to clauses 7.3 and 8.3 of BS 3288.
 - Chemical type tests (composition of materials and galvanizing tests) in accordance to DIN EN 755-7 and ISO 1491.
 - Mechanical type tests (Slip/Tensile strength for Mid-span compression joint) in accordance to clause 7.2 of BS 3288.

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5.2 Sample Tests

- Sample tests shall be carried out on the charge of the contractor in the presence of __ EDC representative at the factory. The sample tests shall include:
 - Conductivity test to verify the requirement of this specification.
 - Verification of dimensions and weights, and check of markings.
 - Uniformity of the final product with the specimen attached to the offer.
- Sample size: Test specimens shall be selected at random from each inspection lot in accordance to ISO 2859-1. The number of selected samples shall be as follows: When the order is for one hundred or more identical fittings, the number of samples for these tests shall be ***p*** or the nearest whole number greater than ***p***, as given by the following formulae, in which ***n*** is the number of fittings in the batch offered for inspection.

$$p = 4 \quad \text{when } 100 \leq n < 500$$

$$p = 4 + \frac{1.5n}{1000} \quad \text{when } 500 \leq n \leq 20000$$

$$p = 19 + \frac{0.75n}{1000} \quad \text{when } n > 20000$$

For batches of fewer than one hundred identical fittings, four samples shall be submitted to the specified non-destructive sample tests.

- In case of failure of test sample to comply with routine tests, __ EDC reserves the right to disqualify the contractor or postpone the delivery until succeeding all the tests. In this case, any delay of supplying shall be on the charge and responsibility of the contractor.

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6- PACKING and MARKING

- All materials shall be packed and delivered in strong and weather resistant boxes of suitable weight to avoid handling problems. Each box shall bear the following information legibly printed in a visible place:
 - Purchase order number / Tender number
 - Accessory catalogue number
 - Manufacturer's name
 - Year of manufacture
 - Quantity of pcs. in the box
 - Gross weight in kilograms (kg)
 - Position of slinging points and other relevant handling instructions

7- GUARANTEE

- The supplier shall guarantee the products against all defects arising out of faulty design or manufacturing defects or defective material for a period of two (2) years from the date of delivery.
- The supplier shall guarantee the uniformity of the products delivered with the approved samples.

8- TRAINING

- The manufacturer / supplier shall provide a sufficient on-site training to __EDC teams for free on the installations and commissioning works.

9- SUBMITTALS

- The Supplier shall furnish detailed dimensioned drawings of each offered item and all its parts. All drawings shall be neatly arranged. All drafting and lettering shall be legible. All dimensions and dimensional tolerances shall be mentioned in mm. All weights shall be mentioned in grams. Drawings shall include the

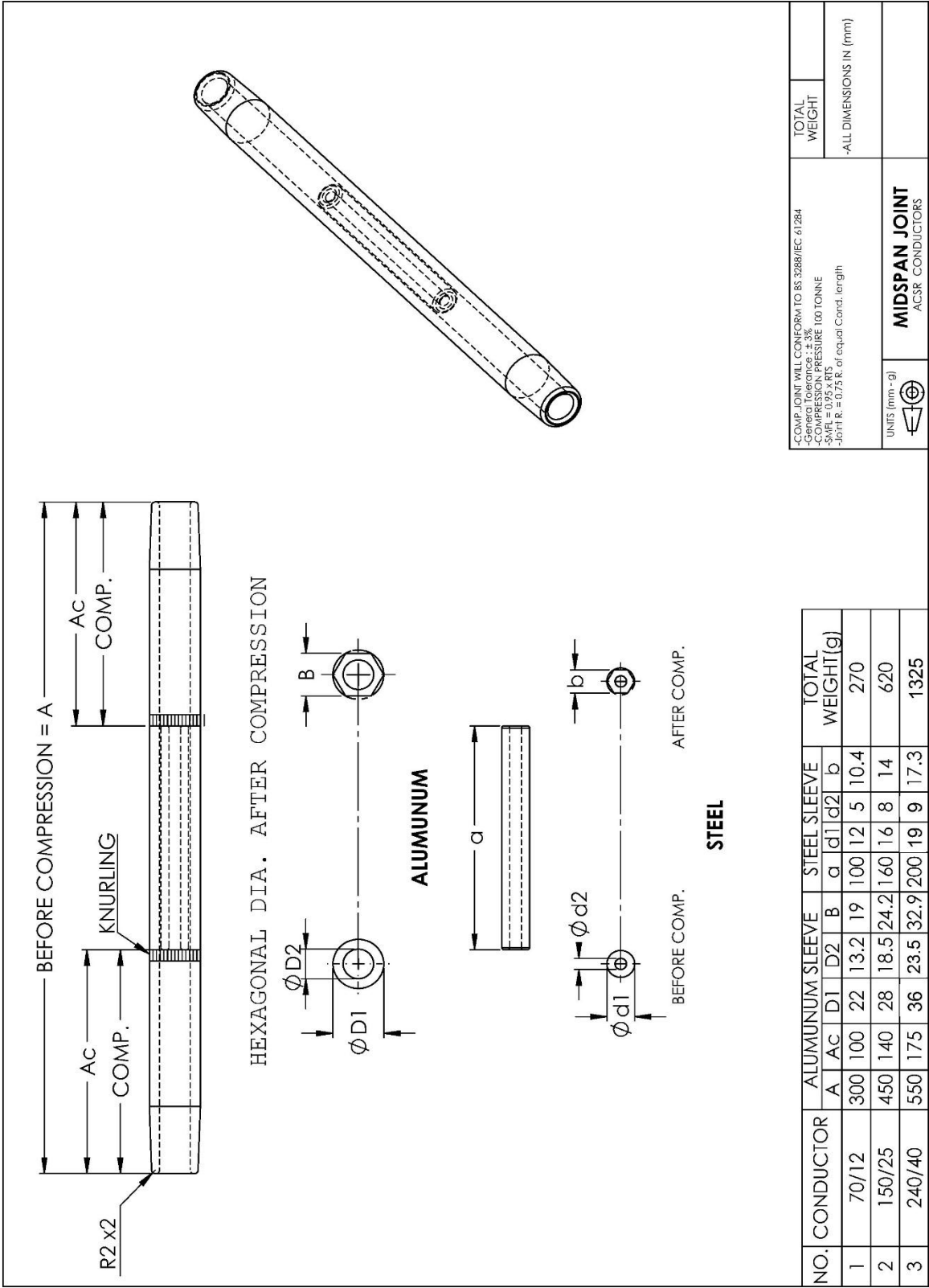
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compression die number with recommended compression pressure and all other relevant technical details

- One specimen of each offered type shall be submitted with the offer for evaluation prior to issuance of the purchase order. Dimensional verification, indelible markings and finishing shall be checked.
- The tenderer must fill in thoroughly the attached technical data schedule for each offered item. Any offer does not accompanied with clear and complete technical data schedule shall be rejected.
- Product type test reports/certificates.
- Approval certificate issued by Egyptian Electricity Holding Company.
- List of deviations and clauses to which exception is taken (if any).

9- DRAWINGS

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BEFORE COMP. AFTER COMP.

ALUMINUM

NO.	CONDUCTOR	A	D1	D2	B	WEIGHT(g)
1	70	300	22	13.2	19	195
2	150	450	28	18.5	24.2	415
3	240	550	36	23.5	32.9	850
4	300	580	38	26	22.5	940

-COMP. JOINT WILL CONFORM TO BS 3286/IEC 61284 -General Tolerance : $\pm 3\%$ -COMPRESSION PRESSURE 100 TONNE -Joint R. = 0.75 R. of equal Cond. length	TOTAL WEIGHT	-ALL DIMENSIONS IN (mm)
UNITS (mm - g)	MIDSPAN JOINT AAAC CONDUCTORS	

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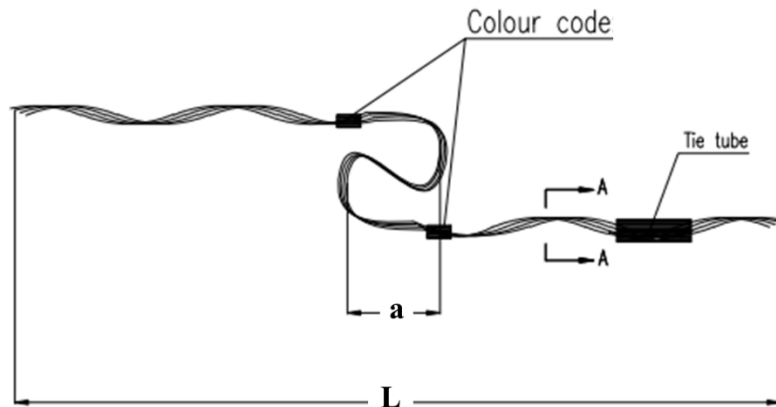
Technical drawings of a parallel groove terminal block. The drawings include a top view showing a rectangular block with two circular contact points, a side view showing the block's profile with a central groove and two side grooves, and a cross-sectional view showing the internal structure with two main contact points and two side contact points. Dimensions are labeled: L (length), W (width), and C (center-to-center distance).

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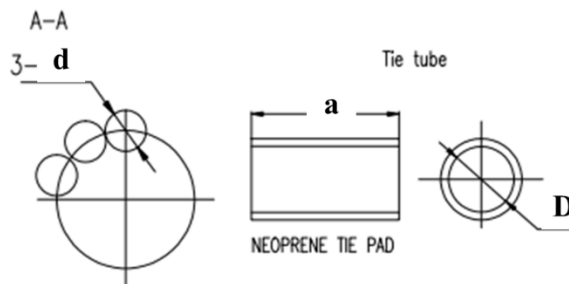
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Top Tie

Preformed Top Tie is made of Aluminum clad steel wire

Handedness: Right



Including Tie pad

Conductor Cross-section mm ²	a Insulator neck dia. min.	D	d rod dia.	L Length	Weight (kg)	Color code
	All dimensions are in mm					
AAAC 70 (dia 10.5)	73	12	3	670	0.15	Green
AAAC 70 (dia 10.5)	89	12	3	670	0.16	Green
AAAC 150 (dia 15.8)	73	16	3.2	690	0.18	Purple
AAAC 150 (dia 15.8)	89	16	3.2	690	0.19	Purple
AAAC 240 (dia 20.3)	73	22	3.2	690	0.19	Yellow
AAAC 240 (dia 20.3)	89	22	3.2	690	0.2	Yellow
AAAC 300 (dia 22.5)	73	24	3.2	690	0.19	Blue
AAAC 300 (dia 22.5)	89	24	3.2	690	0.2	Blue
ACSR 70/12 (dia 11.6)	73	12	3	670	0.15	Red
ACSR 70/12 (dia 11.6)	89	12	3	670	0.16	Red
ACSR 150/25 (dia 17.1)	73	18	3.2	690	0.18	Black
ACSR 150/25 (dia 17.1)	89	18	3.2	690	0.19	Black

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TECHNICAL DATA SCHEDULE

__ EDC Inquiry No: _____ Item No: _____

No.	DESCRIPTION	__ EDC Specified Values	Vendor Proposed Values
	Lugs		
1	Manufacturer	---	
2	Manufacturer Catalogue No.	---	
3	Country of Origin	---	
4	Accessory Type	MS joint / PG Clamp / Top tie	
5	Conductor Type	ACSR / AAAC	
6	Conductor Size (mm ²)	70 – 150 – 240 – 300	
7	Accessory Material Compliance	Yes / No	
8	Accessory Dimension Compliance	Yes / No	
9	Material purity (%)	Al 99.5 %	
10	Joint resistivity	less than 75% of the resistivity of equivalent length of conductor	
11	All ferrous metal parts except those made of stainless steel are protected by hot dip galvanizing	Yes / No	
12	Specimen attached	Yes / No	
13	Submittals Required Included	Yes / No	

I / We guarantee the technical data given above for the offered material/equipment.

Description	Manufacturer of Material/Equipment	Vendor/Supplier
Vendor/Supplier		
Location and Office Address		
Name and Signature of Authorized Representative with Date		
Official Seal / Stamp		